A Structural Selection Framework for Nonlinear Field Systems

Coherence, Entropy, and Instability in Dynamical Field Configurations

Structural Selection & CCI Series (2026)

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Overview

This research series develops a unified framework for understanding which structures persist in nonlinear field systems and why. The central idea is:

> *Not all mathematically valid configurations are dynamically realised. A structural selection principle determines which configurations persist.*

The framework combines:

- nonlinear field dynamics
- maximum entropy selection
- structural free energy
- canonical CCI diagnostics
- dynamical control and benchmark validation

At its core lies the **Critical Coherence Index (CCI)**:

$$\mathcal{C}_{\mathrm{CCI}} = \frac{\text{instability} \times \text{activity}}{\text{coherence}} = \frac{\Gamma_{\mathrm{inst}}\,\Gamma_{\mathrm{prod}}\,(1+\kappa\,U_{\mathrm{struct}})}{\Gamma_{\mathrm{coh}}+\Gamma_{\mathrm{cons}}+\Gamma_{\mathrm{corr}}+\Gamma_{\mathrm{int}}+\varepsilon}$$

Since Paper 23, the CCI is interpreted through a canonical three-level layering:

$$\boxed{\text{formal definition} \longrightarrow \text{physical proxy} \longrightarrow \text{empirical scaling}}$$

$$\mathcal{C}_{\mathrm{CCI}} \approx \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}+\varepsilon} \qquad \text{(field-theory proxy)}$$

$$\boxed{\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \quad \alpha \approx 2.5\ (1\mathrm{D}),\ \alpha \approx 3.0\ (2\mathrm{D}, \text{within tested range})}$$

$$r_s = 0.734\ (1\mathrm{D}), \quad r_s = 0.870\ (2\mathrm{D})$$

The proxy and scaling laws are not the definition of the CCI. In particular, $\dot{S}_{\mathrm{cg}}^{(+)}$ is a coarse-grained histogram-activity proxy in the present Hamiltonian ϕ^4 setting, not a literal thermodynamic entropy production rate.

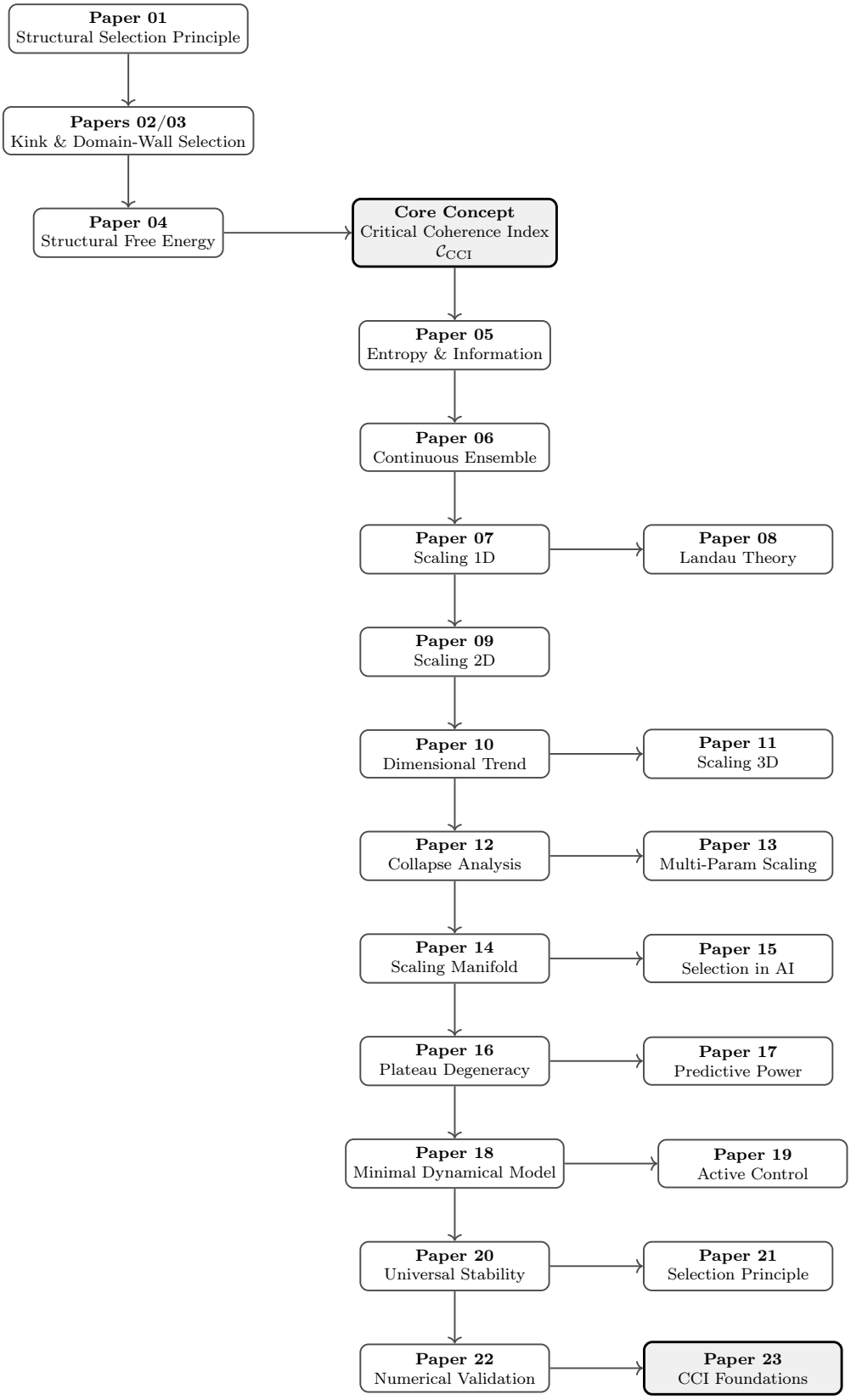
Paper 22 provides the first explicit structural benchmark beyond energy ranking in a minimal ϕ^4 model:

$$\Delta_{\mathrm{vac}} = 2.583852, \qquad p_K = 1.000, \qquad d_F(20\%) = 0.010363 < d_E(20\%) = 0.012128.$$

Across the current series, a four-phase picture emerges:

- **Phase I:** selection principle, diagnostics, and free-energy structure (Papers 01–06)
- **Phase II:** scaling, dimensional dependence, and geometric re-interpretation (Papers 07–16)
- **Phase III:** predictive use, dynamics, control, and principle-level consolidation (Papers 17–21)
- **Phase IV:** benchmark validation and canonical CCI clarification (Papers 22–23)

Paper Series Map



Emergent Picture:
Structure ↔ Information ↔ Stability ↔ Dynamics ↔ Selection

Paper Series and Core Results (01–23)

All papers are available at:

https://kriegchristof.academia.edu/research

1. **Paper 01 — Structural Selection Principle**
   Introduces the core hypothesis: maximum-entropy weighting over physical solution spaces.

$$P[\Phi] \propto \exp(-\beta\, \mathcal{F}_{\text{struct}}[\Phi]), \qquad \mathcal{M}_S = \{\Phi \mid \delta S[\Phi] = 0\}$$

2. **Papers 02/03 — Structural Selection in Nonlinear Field Theories**

   Kink selection in 1D (81/81 settings, weight 99.96%) and domain-wall selection in 2D (transition at $s_\star^c \in (0.20, 0.42)$).

3. **Paper 04 — Structural Free Energy**

   Defines coherence diagnostics, structural free energy, and regime classification via the CCI.

$$\mathcal{F}_{\text{struct}}[\Phi] = \lambda_{\text{stab}}\, E_{\text{stab}} + \lambda_{\text{cons}}\, E_{\text{cons}} + \lambda_{\text{corr}}\, E_{\text{corr}} + \lambda_{\text{dyn}}\, E_{\text{dyn}}$$

4. **Paper 05 — Physical Grounding of the CCI**

   Tests connections between coarse-grained histogram activity and structural information. Spearman $r_s = 0.760$ ($p < 0.0001$) in discrete ensemble.

5. **Paper 06 — Continuous Ensemble Study**

   Strong correlation between CCI and structural free energy in a 160-trajectory ensemble:

$$r_s(\mathcal{C}_{\text{CCI}},\, \mathcal{F}_{\text{struct}}) = 0.878, \quad p < 10^{-50}$$

6. **Paper 07 — Entropy–Information Scaling (1D)**

   Empirical power-law scaling in 1D with phase diagrams:

$$\mathcal{C}_{\text{CCI}} \sim \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha}}, \quad \alpha \approx 2.5, \quad r_s = 0.734, \quad p \approx 1.6 \times 10^{-21}$$

7. **Paper 08 — Landau-Type Theory**
   Proposes a Landau-type theoretical framework for structural selection. Order parameter $\mathcal{O} \sim I_{\text{nn}}$, control parameter $\mu \sim \mathcal{C}_{\text{CCI}}$:

$$\mathcal{F}_{\text{struct}}[\mathcal{O}] = a(\mu)\, \mathcal{O}^2 + b\, \mathcal{O}^4 + \gamma(\nabla \mathcal{O})^2, \quad a(\mu) = a_0(\mu - \mu_c)$$

   Theoretical proposal; Landau coefficients not yet derived from first principles.

8. **Paper 09 — Entropy–Information Scaling (2D)**
   Tests robustness of the scaling in two-dimensional ϕ^4 dynamics:

$$\mathcal{C}_{\text{CCI}} \sim \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha}}, \quad \alpha \approx 3.0 \text{ (within tested range)}, \quad r_s \approx 0.87, \quad R^2 \approx 0.74$$

   Structural information I_{nn} is the dominant factor in 2D; histogram activity is subleading.

9. **Paper 10 — Dimensional Dependence of Scaling**
Comparative analysis of the scaling exponent across 1D and 2D, finding evidence for a dimension-dependent universality class:

$$\alpha_{1D} \approx 2.5 \longrightarrow \alpha_{2D} \gtrsim 3.0, \qquad r_s : 0.734 \to 0.870$$

The dimensional dependence of α is a property of the projection from $\mathcal{F}_{\text{struct}}$ onto the entropy–information plane.

10. **Paper 11 — Entropy–Information Scaling (3D)**
Extension to three-dimensional ϕ^4 dynamics reveals a breakdown of simple power-law scaling. Instead of a sharp exponent, a broad plateau emerges over $\alpha \in [2.8, 3.5]$, with significantly reduced regression power ($R^2 \approx 0.37$) but persistent monotonic correlation:

$$r_s \approx 0.85, \quad p \approx 1.6 \times 10^{-7}, \quad \alpha : \text{plateau } [2.8, 3.5]$$

The notion of a single effective exponent breaks down in 3D — a degeneracy of effective scaling descriptions emerges.

11. **Paper 12 — Cross-Dimensional Collapse Analysis**
Direct visual comparison of the scaling relation across 1D, 2D, and 3D via collapse plots, testing whether a unified cross-dimensional description exists. A partial collapse is observed: the monotonic structure is shared across dimensions, but no single curve provides a universal description.

Scaling survives, but the exponent does not.

12. **Paper 13 — Multi-Parameter Scaling Analysis**
Extends the single-exponent ansatz to a two-parameter model $\mathcal{C}_{\text{CCI}} \sim \dot{S}_{\text{cg}}^{(+)a} I_{\text{nn}}^{-b}$, finding systematic dimensional restructuring of the scaling weights:

$$\begin{aligned} &\text{1D: } a \approx -1.6,\ b \approx 0.33,\ R^2 \approx 0.76 \\ &\text{2D: } a \approx -0.08,\ b \approx 0.50,\ R^2 \approx 0.74 \\ &\text{3D: } a \approx +0.30,\ b \approx 0.30,\ R^2 \approx 0.37 \end{aligned}$$

The entropy weight a changes sign from 1D to 3D, revealing a qualitative shift in the role of dynamical activity. *The exponent description fails not because the scaling breaks, but because the system transitions to a higher-dimensional multi-parameter scaling structure.*

13. **Paper 14 — Scaling Manifolds in Nonlinear Field Systems**
Interprets the observed scaling laws as projections of a higher-dimensional manifold,

$$\mathcal{M}_{\text{scale}} = \{(\dot{S}_{\text{cg}}^{(+)}, I_{\text{nn}}, d, \mathcal{C}_{\text{CCI}})\},$$

with empirical evidence that anisotropy and exponent instability are linked. *Observed power laws are treated as lower-dimensional views of a more general geometric organisation.*

14. **Paper 15 — Structural Selection in Artificial Intelligence**
Extends the structural-selection perspective beyond field systems:

$$\text{learning} \sim \text{structural selection.}$$

The paper is conceptual rather than derivational, but establishes the bridge from scaling manifolds to representation selection and constraint-aware AI.

15. **Paper 16 — Plateau Degeneracy in Entropy–Information Scaling**
Introduces a quantitative description of exponent flatness and scaling breakdown, with the strongest degeneracy occurring in 3D. *The breakdown of one-parameter scaling is itself measurable and dimension-sensitive.*

16. **Paper 17 — Predictive Power of the CCI**
Tests the CCI as a regime classifier:

$$\text{accuracy} = 1.000, \qquad \bar{A}_{\text{CV}} = 0.992 \pm 0.019.$$

Ordered, critical, and chaotic states can be separated by the CCI alone to very high accuracy in the tested ensembles.

17. **Paper 18 — A Minimal Dynamical Model for Structural Selection**
Introduces a coarse-grained dynamical field equation for coherence:

$$\partial_t C = D\nabla^2 C + aC - bC^3.$$

This gives the first dynamical realisation of structural selection beyond purely ensemble-based observation.

18. **Paper 19 — Active Structural Control in Nonlinear Field Systems**
Extends the passive dynamical model by feedback control:

$$\partial_t C = D\nabla^2 C + aC - bC^3 + \lambda U, \qquad U = -(C - C^*).$$

Numerically, the strongest control response occurs in the low-damping regime, with a qualitative crossover near $\gamma_c \approx 0.03$–0.04.

19. **Paper 20 — Universal Stability of Structural Observables Across Dimensions**
Shows that the reduced observable $\mathcal{C}_{\text{CCI}}$ and the higher-level functional $\mathcal{F}_{\text{struct}}$ are comparatively dimension-stable, whereas the raw entropy– information ratio is much less stable. *This supports the view that the proxy is not the object.*

20. **Paper 21 — Structural Coherence as a Selection Principle**
Provides the principle-level synthesis:

$$\Phi_{\text{selected}} = \arg\min_{\Phi} \mathcal{F}_{\text{struct}}[\Phi], \qquad \partial_t C = -\frac{\delta \mathcal{F}}{\delta C}, \qquad \frac{d\mathcal{F}}{dt} \leq 0.$$

Structural coherence is elevated from a descriptive motif to a phenomenological selection principle.

21. **Paper 22 — Numerical Validation of Structural Selection**
Presents the first explicit minimal ϕ^4 benchmark in which a structural ranking is shown to be operational and not reducible to static energy alone. Three quantitative outcomes are central:

$$\Delta_{\text{vac}} = 2.583852, \qquad p_K = 1.000, \qquad d_F(20\%) = 0.010363 < d_E(20\%) = 0.012128.$$

Structural ranking matches energy at the sharpest percentiles and improves upon it at the 20% level.

22. **Paper 23 — Structural Foundations of the Critical Coherence Index**
Canonises the CCI by separating its three theoretical layers:

$$\text{definition} \longrightarrow \text{proxy} \longrightarrow \text{scaling}.$$

For nonlinear field systems, the Γ sectors are closed explicitly, and for cross-domain transport a universal closure rule is introduced:

$$\Gamma_a^{(D)}[x] = \mathcal{N}_{\sigma_a^{(D)}}\left(r_a^{(D)}[x]; \Lambda_a^{(D)}\right).$$

This paper is a clarification note rather than a new benchmark, but it stabilises the meaning of the CCI across the full series.

Hierarchical Interpretation

A coherent three-level picture emerges across the series:

Level	Quantity	Role
Formal observable	$\mathcal{C}_{\mathrm{CCI}}$	Canonical structural instability index
Physical proxy	$\dot{S}_{\mathrm{cg}}^{(+)}/(I_{\mathrm{nn}} + \varepsilon)$	Measurable field-theory representation
Empirical scaling	$\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$	Ensemble-level scaling description
Selection functional	$\mathcal{F}_{\mathrm{struct}}$	Global effective selection landscape

The resulting high-level hierarchy may be summarised as:

$$\text{diagnostics} \longrightarrow \mathcal{C}_{\mathrm{CCI}} \longrightarrow \mathcal{F}_{\mathrm{struct}}.$$

And the three dynamical regimes remain:

Regime	I_{nn}	$\dot{S}_{\mathrm{cg}}^{(+)}$
Ordered	high	low
Critical	moderate	moderate
Chaotic	low	high

The CCI plays a role analogous to Reynolds-type numbers in fluid and plasma systems: low CCI preserves coherent structures; high CCI drives the system toward chaotic, instability-dominated behaviour.

Future Work

- Extension of the benchmark programme beyond the minimal ϕ^4 setting
- Validation of the cross-domain closure schema introduced in Paper 23
- Theoretical derivation of the multi-parameter scaling family $\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{(+)a}\, I_{\mathrm{nn}}^{-b}$ from $\mathcal{F}_{\mathrm{struct}}$
- Application to other nonlinear models, including sine-Gordon, Kuramoto–Sivashinsky, and reaction–diffusion
- Sharper connection between active control, CCI reduction, and free-energy descent
- First-principles derivation of the CCI sectors and of their dimensional dependence

Code & Data

All companion code is openly available:

https://github.com/Chris4081/structural-selection-principle

The repository now contains:

- simulation and analysis scripts for the 1D, 2D, and 3D scaling studies
- control and benchmark scripts for Papers 19 and 22
- figure-generation pipelines used in the current paper series
- canonical CCI and benchmark source files accompanying Papers 22 and 23